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Performance Analysis of Data-oriented Programming in Rust



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Abstract

< Abstract in english >

Kurzfassung

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Chapter 1

Introduction

1.1 Motivation

In the past decades, processor speeds have vastly increased due to increased clock frequency and higher core counts, while DRAM has not been able to keep up. This phenomenon is colloquially known as the "memory wall". The resulting problem is that execution speed may no longer be limited by processing power, but by the processor having to wait for data from the DRAM [5]. To address this gap, modern software development has to take into consideration the underlying hardware constraints.

1.2 The Shift to DOP

Object-oriented Programming (OOP) has enjoyed wide adoption and is usually taught long before Data-oriented Programming (DOP) in most curricula. However, OOP poses challenges when it comes to cache-optimization, for example, due to objects bundling data and logic, which leads to inefficient memory layouts and cache pollution [7]. More modern approaches, such as DOP, aim to mitigate challenges such as these by decoupling logic and data. However, there are more mechanisms that DOP employs to improve cache efficiency and execution time, some of the more significant ones being Single Instruction, Multiple Data (SIMD) vectorization (Section 2.3) and use of the Entity Component System (ECS) software architecture design pattern (Section 2.4).

1.3 Objectives and Methodology

The aim of this thesis is to measure the difference in execution time between OOP and DOP approaches, as the exact difference is hardly ever quantified. The created benchmarks aim to reflect realistic use cases and lean towards the gaming industry, as there has been a significant push in recent years to adopt the ECS pattern. Rust will act as the programming language with the Bevy engine providing the framework. Rust provides memory safety without requiring a garbage collector [10]. The absence of garbage collection is especially desirable for benchmarking, as it eliminates nondeterministic collection pauses that would otherwise distort time measurements. Bevy builds directly on these properties. According to its documentation, Bevy ECS represents components as plain Rust structs and systems as ordinary functions, which the project contrasts with other Rust ECS implementations that often rely on more complex constructs such as lifetimes or builder patterns [2]. This keeps the framework-based implementation close in form to the hand-written architectures it is compared against, which is desirable when the comparison is meant to isolate memory layout rather than API design.

Chapter 2

Background

This section describes the relevant parts of DOP, which also include SIMD vectorization and the ECS design pattern, as they are integral to the nature of the systems under inspection. Furthermore, a brief overview of Rust and the Bevy engine is presented.

2.1 Data-oriented Programming

The OOP paradigm undoubtedly has many useful features such as encapsulation, polymorphism and inheritance. These features enable software developers to think of problems in terms of objects and their interactions, with objects ideally being self-contained. Due to this, programmers are able to easily reuse and maintain them without having to fear side effects [1]. Nevertheless, objects encapsulating the entirety of their data and logic is a double-edged sword.

For example, a program that solely updates the position of some objects, where each object also holds further attributes, e.g. color or weight. Due to encapsulation, the CPU has to load all the data of these objects into its cache, even though it should only have to load the location for its computation. This phenomenon is often called "cache pollution" [5] which poses a problem due to another phenomenon called the "memory wall", which describes the growing gap between processor and memory speeds: as processor speed increases faster than memory speed, an increasing number of programs becomes limited not by clock frequency but by the time spent waiting for data from memory. Over the past decades, processor performance has improved far more quickly than memory speed. Wulf and McKee [15] illustrated this disparity in 1995, reporting an annual increase of roughly 80 % in processor speed against only 7 % for memory. While the exact rates have changed since, the underlying trend has persisted, and the resulting gap has become one

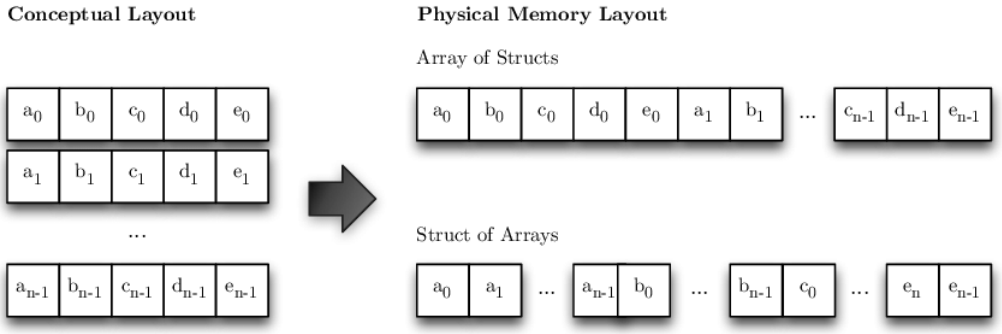


Figure 2.1: Memory Layout Visualization: AoS vs. SoA (taken from [12])

of the biggest bottlenecks in high-performance computing, as the CPU often stays idle while waiting for data to be fetched from main memory [9].

One of the most promising solutions to these problems is DOP, as it aims to decouple data from logic and thus reduce cache pollution and the CPU idling [6]. DOP improves CPU utilization by fundamentally changing how data is handled by moving away from the classical OOP memory layout called Array of Structures (AoS) and instead using a more optimized Structure of Arrays (SoA) layout [6].

In Figure 2.1, both memory layouts are compared. On the conceptual level, there are simply structs with data, e.g. objects with fields. The more conventional AoS layout stores these structs sequentially within an array, thus the name. In contrast, the SoA layout creates multiple arrays which contain only one type of field. As an example, one could want to manage 1000 `Person` objects which possess the fields `String name`, `float weight` and `Color hairColor`. The AoS layout would result in a `Person` Array with 1000 entries. Contrarily, the SoA layout would result in three arrays, one of type `String` containing the `name` values of all 1000 `Person` instances. Similarly, an array is created for `weight` and `hairColor`.

This is due to CPU caches exploiting a principle called spatial locality. Drepper argues [5] that since neighboring data is likely to be used together, it should be loaded into the cache together. This is what the entire idea of SoA is based upon. If one wants to perform a read or write operation on only the `name` of these 1000 `Person` instances, the CPU can load much more relevant data than with the AoS layout, as the cache is not clogged by the irrelevant `weight` and `hairColor` data. Thus, while the SoA layout might appear confusing to programmers used to the AoS layout, it is necessary in order to properly achieve the performance gains granted by SIMD vectorization.

2.2 Caching and CPU-Architecture

2.3 Data-Level Parallelism via SIMD

Traditionally, to perform operations on data, the CPU has to load these values into a General Purpose Register (GPR). These usually have a fixed size of 32-bit or 64-bit, meaning an instruction can operate on two values of this size at once. For example, assuming a 32-bit environment, a scalar add instruction can add no more than two 32-bit values per cycle.

In order to overcome this limitation, modern CPUs contain specialized SIMD vector units. These units use much wider registers, such as the 512-bit registers found in the Intel MIC architecture, which allow the CPU to treat a single register as a vector of values. By using 512-bit registers, operations can be performed on up to 16 single-precision or 8 double-precision floating-point values in parallel [14]. On consumer grade CPUs, the AVX2 instruction set is widely adopted, using 256-bit registers.

This effectively parallelizes the computation and significantly increases throughput, under the condition that the data is laid out in memory as explained in Section 2.1. When looking at the memory layouts in Figure 2.1 it becomes clear why SIMD vectorization can only bring significant performance gains when the data is stored in a SoA layout. If the data was instead organized as in the AoS layout, SIMD units could not be fed with packed, homogenous data, causing the cache to be filled with irrelevant data that hinders the aspired parallel execution of operations. To achieve the outlined hardware efficiencies

without the programmer having to manually manage memory, the ECS design pattern may be employed.

2.4 Entity Component Systems

As the presentation by Bilas [4] at the Game Developers Conference in 2002 shows, there has been a desperate need for the ECS pattern since at least the early 2000s. He explains that with increasing complexity of game systems, the resulting hardcoded class tree must undergo repeated refactoring and will naturally stray further and further from clean OOP principles. He noticed that when re-examining the problem of complex object schemes or structures, that he could also view it as a database and that in the end, only the data matters.

This data-centric perspective is the essence of the ECS pattern. It encourages clean separation of data and logic by employing a 3-level architecture as can be seen in Figure 2.2. (1) Entities can be viewed as an identifier value to which components may be attributed. (2) A Component is a domain-specific datatype and its value is comparable to the field value of an object. One entity may possess multiple components. (3) Lastly, a system is a domain-specific piece of logic or behavior. This system performs computations on entities and their components. Entities and Components represent a game state, with the Systems performing the transitions between states [13].

As an example, we revisit the `Person` objects from Section 2.1. We discussed that the SoA memory layout would greatly improve performance, but we did not explain how this layout would be achieved. Traditionally, one would define a `Person` class with fields and methods and then instantiate objects. To transform this into a SoA memory layout, the ECS pattern is applied as follows:

- Entities are created as unique identifiers.
- Components such as `name`, `weight` and `hairColor` are associated with these entities. In an archetype based ECS like Bevy, this ensures that all `weight` components are stored in a contiguous array, separate from `hairColor` [3].
- Lastly, Systems are defined to perform transformations by querying for entities that possess a certain set of components. For example, updating the `hairColor` component of all `Person` instances that possess the `name` and `hairColor` components. This allows the system to iterate over a tightly packed array of only relevant data, enabling the use of SIMD instructions.

2.5 Rust and Bevy engine

Rust is a systems programming language which provides performance comparable to C++ and memory safety through its unique ownership system. The compiler enforces strict rules on how data may be accessed and modified at compile-time. By doing so, Rust eliminates entire classes of bugs such as data races, and without requiring a garbage collector [10].

The Bevy engine expands upon this foundation, using an archetype-based ECS framework that prioritizes performance and modularity. Archetypes group entities with identical combinations of components, which enforces the SoA memory layout [3].

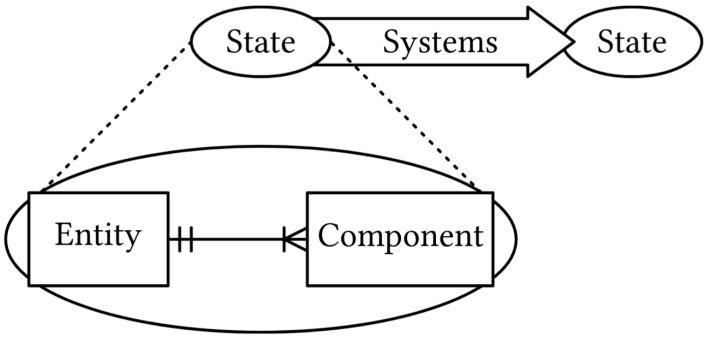


Figure 2.2: Visualization of the ECS pattern depicting (1) Entities, (2) Components and (3) Systems (taken from [13])

Chapter 3

Benchmark Design and Implementation

3.1 Simulation

To properly analyze the differences in performance between DOP and OOP, the simulation used must be conceptually and mathematically identical across all evaluated architectures. Otherwise, differences in performance cannot be reliably attributed to the effects of adhering to one paradigm or the other, as they could just as likely occur due to inhomogeneous workloads. Thus, all three scenarios will perform calculations and operations on the same core simulation. Additionally, the simulation must provide a workload that is non-trivial enough to cause the property under inspection to dominate the runtime, as well as enable all relevant properties to be inspected. For example, while a complex calculation may be non-trivial enough to dominate the runtime in a compute-bound scenario, it does not exercise the memory hierarchy sufficiently to inspect the performance differences of different memory layouts.

The benchmark simulation performs gravitational calculations on a certain number of planets. These planets always have the same three conceptual attributes, a (1) 3D position in space, a (2) 3D velocity and (3) a payload that acts as a way to artificially inflate the size of the planet objects. The position and velocity of all planets are randomly generated during the setup phase with a seeded random number generator, while the payload is a known and fixed size at compile time. This design enables all three scenarios to work with the same underlying simulation. In Scenario 1, the gravitational calculation dominates the runtime. In Scenario 2, the payload is changed iteratively to inspect the performance differences in memory layouts. Lastly, in Scenario 3, an arbitrary number of planets can be generated, allocated and then deallocated to simulate a particle system.

Regarding the calculation of gravitational pull, the simulation assumes that all planets are accelerated towards a central body of a large fixed mass (e.g. a black hole). Due to the planets being accelerated towards this central mass, the acceleration of a planet depends only on its position relative to that mass, not on its own mass: by Newton's second law the planet's mass cancels out. The mass of the central body is constant and is folded into the gravitational constant. The acceleration therefore simplifies to

$$\vec{a} = \frac{G}{r^2} \cdot \hat{d} \tag{3.1}$$

where r is the distance between the planet and the centre, $\hat{d}$ is the unit vector pointing from the planet towards the centre, and G subsumes both the gravitational constant and the (constant) central mass. The planet's velocity and position are then integrated over a fixed time step Δt.

3.2 Architectures Evaluated

Chapter 2 contrasted OOP and DOP on the level of paradigms. However, a paradigm itself cannot be benchmarked. Concepts such as encapsulation or a data-oriented mindset leave no direct trace in hardware. What does reach the hardware is the resulting arrangement of data in memory and the pattern in which that data is accessed, which can then be measured. Thus, the comparison is reduced to representative implementations that differ precisely in these two properties. Since the simulated workload itself is identical across all of them (Section 3.1), any measured difference can be attributed to the layout rather than to the workload. In total, four architectures are evaluated, two of which represent OOP and two of which represent DOP design principles. They are arranged so that each of them differs from the preceding one in primarily one property, spanning a range from maximum indirection to maximum contiguity.

Pointer Chasing (PC) The PC architecture stores planets as `Vec<Box<PlanetAoS>>`. Every planet is allocated individually on the heap, with the vector solely holding the pointers to said planets. In Rust, this indirection is provided by `Box<T>`, where the content of the box lives on the heap and the box itself is a pointer of known size referring to the content [10]. This architecture represents the reference semantics of common object-oriented languages such as Java or C#, where objects are allocated individually and accessed through references.

Since caches exploit spatial locality [5], this layout is the least favorable one under inspection, as traversing the collection means following a pointer per element instead of having the data be laid out in a contiguous manner. This traversal pattern is commonly referred to as pointer chasing, and Drepper employs it in the same form, as a randomly ordered linked structure, to demonstrate the limits of hardware prefetching [5]. However, when objects are created within a single uninterrupted loop, a general-purpose allocator may place them in near-contiguous memory, in which case the indirection under inspection would not actually occur. To combat this, the vector is shuffled once after setup. This is also the more faithful representation of a long-living OOP application, whose heap has become fragmented over time. Without it, the architecture would degenerate into the AoS architecture described below and likely understate the performance differences.

Array of Structures (AoS) The AoS architecture stores the identical `PlanetAoS` struct in a `Vec<PlanetAoS>`, i.e. without any indirection. The planets are laid out contiguously within the vector, whereas the fields of a single planet remain interleaved: position, velocity and payload of one planet are stored adjacently, followed by the fields of the next planet.

This architecture therefore represents the best case an object-oriented layout can achieve. The indirection present in the PC architecture is removed while retaining the conceptual properties of an object, namely that all of its data is kept together. As the AoS and PC architectures share the exact same `PlanetAoS` struct and differ solely in the presence of a `Box<T>`, any measured difference may be directly attributed to the indirection alone.

Structure of Arrays (SoA) The SoA architecture, in contrast to the previous two, adheres to DOP principles. Instead of one vector holding whole planets, one vector is created per individual field. This results in three vectors for the x, y and z coordinates of a planet's position and in an additional three vectors for the x, y and z values of a planet's velocity. Together with a seventh vector holding the payloads, a planet is thus

no longer a singular entity with inhomogeneous data, but an index shared across seven homogeneous vectors.

This is the layout as described in Section 2.1 and represents the data-oriented ideal. It is implemented manually without any framework, so that it forms a lower bound. Any performance costs incurred by abstraction built on top of a SoA layout are measured in relation to this baseline. However, manually managing and synchronizing seven parallel vectors is error-prone and cumbersome and does not reflect how data-oriented code is written in practice.

Bevy ECS The Bevy architecture therefore represents the same idea as the SoA architecture, but implemented with an ECS framework. Bevy groups entities (in this case planets) by archetypes, i.e. by their exact set of components [13], and stores the components of an archetype in contiguous, per-component storages [3]. Since all planets in this simulation possess an identical combination of components (`Position`, `Velocity`, `Payload`), they occupy the same archetype and the resulting storage pattern is thus conceptually equivalent to that of the SoA architecture.

The difference to the SoA architecture is therefore not the way the data is stored, but the surrounding abstraction required to make such a data-oriented layout maintainable. This includes entity identifiers, archetype metadata and especially the query mechanism through which components are accessed. Ultimately, Bevy represents DOP as it is realistically adopted and the comparison with the SoA architecture quantifies how much that convenience costs.

Vector Types and Comparability The architectures do not only differ in memory layout, but also in their representation of position and velocity. While PC and AoS store them as `Vec3`, i.e. three consecutive `f32` values, the SoA architecture stores the individual coordinates in separate vectors of `f32`. The Bevy architecture uses `Vec3A`, which extends the three `f32` values by four bytes of padding and is aligned to a 16-byte boundary [8]. A single position or velocity value therefore occupies 16 bytes instead of the 12 bytes occupied by the three coordinates in the remaining architectures. This is a deliberate decision, as each architecture uses the representation that is idiomatic for it. The measurements thus reflect the performance of code as it would be written in practice, instead of code that has been homogenized for the sake of uniformity.

Enforcing a single representation would distort the comparison in one direction or the other. Forcing Bevy to use `Vec3` would deny it the aligned accesses that vectorized instructions rely on, since 16-byte memory accesses are expected to be 16-byte aligned and the instruction variants handling unaligned accesses are slower [5]. Conversely, using `Vec3A` throughout would add padding to the object-oriented structs and thereby inflate the size of the very objects whose size is under inspection.

The consequence of this decision is that the amount of hot data per planet is not identical across all architectures. Except for Bevy, which carries 32 bytes of position and velocity per planet, all architectures carry 24 bytes. Bevy therefore trades a larger data footprint for the alignment its layout exploits. The payload, in contrast, is identical across all architectures and, when varied, is varied for all four simultaneously. This is relevant insofar as it is one of the parameters varied in Scenarios 2 and 3.

3.3 Scenario 1: Compute-Bound

- Setup and Update Logic, SIMD

3.4 Scenario 2: Memory-Bound

- Setup and Update Logic, Cache Pollution / Memory Wall

3.5 Scenario 3: Allocation and Deallocation

- Setup and Update Logic, Drop vs Reuse

3.6 Execution and Configuration

All architectures and benchmarks are compiled into a single executable located at `../Benchmarks/target/`. An individual benchmark run is configured almost entirely through Command Line Interface (CLI) arguments, with the parsing of the commands being handled by the `clap` crate [11]. Table 3.1 lists the available arguments.

Short	Long	Default	Description
`-b`	`--benchmark`	required	Scenario: `s1`, `s2`, `s3` or `reset`
`-p`	`--planets`	10 000	Number of planets created during setup
`-w`	`--warmups`	2500	Warmup iterations, results discarded
`-i`	`--iterations`	5000	Measured iterations per sample
`-s`	`--samples`	5	Number of samples per invocation
`-c`	`--chunksize`	100	Elements added and removed per iteration (S3)
`-r`	`--reuse`	`false`	Enables the reuse variant (S3)
`-d`	`--debug`	`false`	Prints a runtime summary to the console

Table 3.1: Command-line arguments for configuring the benchmark executable

The program itself does not contain the functionality of parameter sweeping, i.e. varying parameter values within defined ranges. Instead, PowerShell and bash scripts are provided which repeatedly execute the benchmark with varying values. This keeps the measured code free from any sweeping logic and makes every measurement reproducible from its arguments alone. In addition, the scripts recompile the benchmarks with varying payload sizes for Scenarios 2 and 3, as the payload size is a compile-time constant and thus must be defined before the execution of the benchmarks.

A single invocation executes all four architectures in sequence. For every architecture and for every sample, the corresponding benchmark function is called twice: once with the number of warmup iterations and once with the number of benchmark iterations. Calling the same function for both warmup and benchmark iterations ensures that the cache is in a comparable state when the measurements begin. A typical invocation is shown below.

```
cargo run --profile o3 -- -b s3 -w 1000 -i 5000 -s 5 -c 500 -r -d
```

Measurement results are appended to a Comma-Separated Values (CSV) file per architecture and scenario. For example, the above invocation would append the measurements to the files `bevy.csv, soa.csv, etc.` located at `../Benchmarks/diagrams/s3/csv/`. Each line contains an entry for the respective architecture, planet count, number of iterations, chunk size, payload size in bytes as well as the elapsed time in nanoseconds. In order to delete the generated CSV files, the `reset` value for the `benchmark` argument may be used. This acts as a way to create a clean slate and prevent appending measurements to those of previous benchmarks, should the CSV files not have been moved yet.

Chapter 4

Evaluation

4.1 Methodology

- Benchmark Setup incl. Compiler Flags
- Hardware/System Description incl. CPU values

4.2 Scenario 1: Compute-Bound

- Results and Analysis
- SIMD Details

4.3 Scenario 2: Memory-Bound

- Results and Analysis
- "Anomalies", Cache

4.4 Scenario 3: Allocation and Deallocation

- Results and Analysis
- drop vs reuse

4.5 Discussion

- High Level Overview and Comparison of findings

4.6 Limitations

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Chapter 6

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List of Acronyms

OOP Object-oriented Programming

DOP Data-oriented Programming

PC Pointer Chasing

AoS Array of Structures

SoA Structure of Arrays

GPR General Purpose Register

SIMD Single Instruction, Multiple Data

ECS Entity Component System

CLI Command Line Interface

CSV Comma-Separated Values